05-Metrics-Guide

Purple Team Metrics Guide

Metric Categories

1. Detection Strategy Coverage

Question: Can we see the activity?

Metric	Description	Target
Local Logging	Events recorded on endpoint	Yes for all TTPs
Central Logging	Events forwarded to SIEM	Yes for all TTPs
Detection Strategy Coverage	% of ATT&CK detection strategies implemented	>80%

Tool: DeTT&CT for detection coverage scoring against ATT&CK v18

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2. Detection Fidelity

Question: Do we get notified, and at what level of detail?

Level	Description	Example
None	No visibility	Blind spot
Telemetry	Raw logs available	Process creation logged
IOC Match	Known-bad indicator hit	Hash match
General Behavior	Suspicious pattern flagged	Unusual process tree
Specific Behavior	Exact technique identified	"Credential dumping via LSASS"

Scoring per TTP:

Score	Detection Level
0	None
1	Telemetry only
2	IOC-based detection
3	General behavior alert
4	Specific technique alert
5	Full attack chain correlated

Note: This scale measures detection quality (what type of detection fired). For measuring the full detection-to-response outcome, use the Graded Outcome Scale (0-5) in the ATT&CK Navigator Cheat Sheet.

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3. Response Timing

Question: How fast do we respond?

Metric	Definition	Measurement
TTD (Time to Detect)	Execution -> First detection	Minutes
TTI (Time to Investigate)	Detection -> Scope understood	Minutes
TTR (Time to Remediate)	Scope -> Contained/Eradicated	Minutes/Hours

```
[Execution] --TTD--> [Detection] --TTI--> [Scoped] --TTR--> [Remediated]
```

TTP Tracking Template

#	TTP	ATT&CK ID	Detection Level	TTD	TTI	TTR	Gaps	Action Items
1			0-5	min	min	min		
2			0-5	min	min	min		
3			0-5	min	min	min		

Aggregated Metrics

Exercise Summary

Metric	Formula	Target
Detection Rate	(TTPs Detected / Total TTPs) x 100	>70%
Avg Detection Level	Sum of scores / Total TTPs	>3.0
Avg TTD	Sum of TTD / Detected TTPs	<15 min
Avg TTI	Sum of TTI / Detected TTPs	<30 min
Coverage Gaps	TTPs with score 0-1	<20%

Trend Tracking (Across Exercises)

Exercise	Date	TTPs Tested	Detection Rate	Avg TTD	Improvements
PT-001			%	min	
PT-002			%	min	
PT-003			%	min	

Detection Maturity Progression

```
 Level 1: VISIBILITY
├── Can we see it? (Detection strategies)
│
Level 2: DETECTION
├── Do we get alerted? (Rules/Analytics)
│
Level 3: CORRELATION
├── Can we connect the dots? (Attack chain)
│
Level 4: RESPONSE
├── How fast can we act? (TTD/TTI/TTR)
```

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Level 5: ENGINEERING
└── Can we sustain it? (Detection-as-Code, regression testing)
```

Priority: Build visibility before detection before response measurement

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Action Item Categories

Category	Description	Owner
Detection Coverage Gap	Need to implement additional detection strategies	IT/Security Engineering
Detection Gap	Need new rule/alert	Detection Engineering
Process Gap	Need updated playbook/procedure	SOC/Hunt/DFIR
Tool Gap	Need new capability	Security Architecture
Training Gap	Need analyst skill development	Training/Management

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Reporting Template

Executive Summary

- Total TTPs tested: X
- Detection rate: X%
- Average time to detect: X min
- Critical gaps identified: X
- Improvements since last exercise: X

Key Findings

1. [Finding 1]
2. [Finding 2]
3. [Finding 3]

Action Items by Priority

Priority	Item	Owner	Due Date
High			
Medium			
Low			
